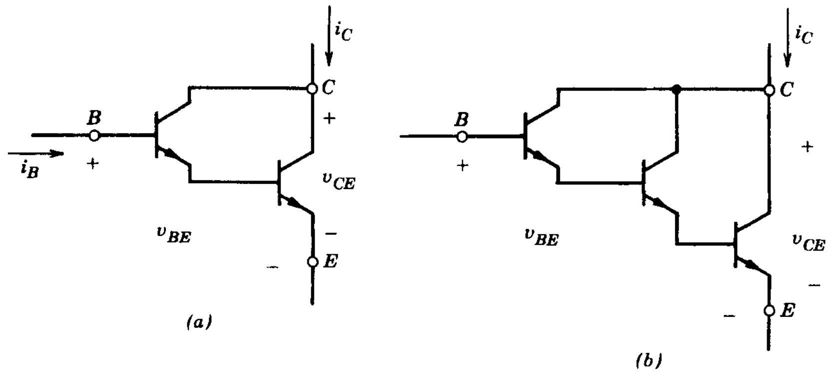

Figure 2-8 Darlington configurations: (*a*) Darlington, (*b*) triple Darlington.

Including MDs, BJTs are available in voltage ratings up to 1400 V and current ratings of a few hundred amperes. In spite of a negative temperature coefficient of on-state resistance, modern BJTs fabricated with good quality control can be paralleled provided that care is taken in the circuit layout and that some extra current margin is provided, that is, where theoretically four transistors in parallel would suffice based on equal current sharing, five may be used to tolerate a slight current imbalance.

2-6 METAL-OXIDE-SEMICONDUCTOR FIELD EFFECT TRANSISTORS

The circuit symbol of an *n*-channel MOSFET is shown in Fig. 2-9*a*. It is a voltage-controlled device, as is indicated by the *i–v* characteristics shown in Fig. 2-9*b*. The device is fully on and approximates a closed switch when the gate–source voltage is below the threshold value, $V_{GS(\text{th})}$. The idealized characteristics of the device operating as a switch are shown in Fig. 2-9*c*.

Metal–oxide–semiconductor field effect transistors require the continuous application of a gate–source voltage of appropriate magnitude in order to be in the on state. No gate current flows except during the transitions from on to off or vice versa when the gate capacitance is being charged or discharged. The switching times are very short, being in

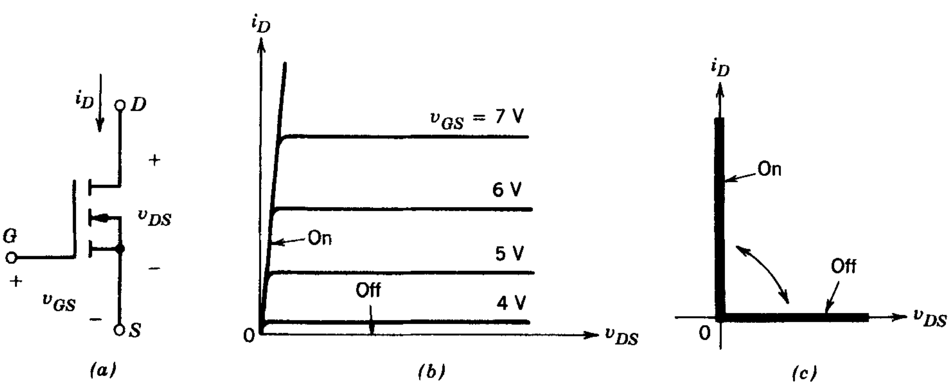

Figure 2-9 *N*-channel MOSFET: (*a*) symbol, (*b*) *i–v* characteristics, (*c*) idealized characteristics.

the range of a few tens of nanoseconds to a few hundred nanoseconds depending on the device type.

The on-state resistance $r_{DS(\text{on})}$ of the MOSFET between the drain and source increases rapidly with the device blocking voltage rating. On a per-unit area basis, the on-state resistance as a function of blocking voltage rating BV_{DSS} can be expressed as

$$r_{DS(\text{on})} = k\, BV_{DSS}^{2.5-2.7} \tag{2-9}$$

where k is a constant that depends on the device geometry. Because of this, only devices with small voltage ratings are available that have low on-state resistance and hence small conduction losses.

However, because of their fast switching speed, the switching losses can be small in accordance with Eq. 2-6. From a total power loss standpoint, 300–400-V MOSFETs compete with bipolar transistors only if the switching frequency is in excess of 30–100 kHz. However, no definite statement can be made about the crossover frequency because it depends on the operating voltages, with low voltages favoring the MOSFET.

Metal–oxide–semiconductor field effect transistors are available in voltage ratings in excess of 1000 V but with small current ratings and with up to 100 A at small voltage ratings. The maximum gate–source voltage is ±20 V, although MOSFETs that can be controlled by 5-V signals are available.

Because their on-state resistance has a positive temperature coefficient, MOSFETs are easily paralleled. This causes the device conducting the higher current to heat up and thus forces it to equitably share its current with the other MOSFETs in parallel.

2-7 GATE-TURN-OFF THYRISTORS

The circuit symbol for the GTO is shown in Fig. 2-10*a* and its steady-state *i–v* characteristic is shown in Fig. 2-10*b*.

Like the thyristor, the GTO can be turned on by a short-duration gate current pulse, and once in the on-state, the GTO may stay on without any further gate current. However, unlike the thyristor, the GTO can be turned off by applying a negative gate-cathode voltage, therefore causing a sufficiently large negative gate current to flow. This negative gate current need only flow for a few microseconds (during the turn-off time), but it must have a very large magnitude, typically as large as one-third the anode current being turned off. The GTOs can block negative voltages whose magnitude depends on the details of the

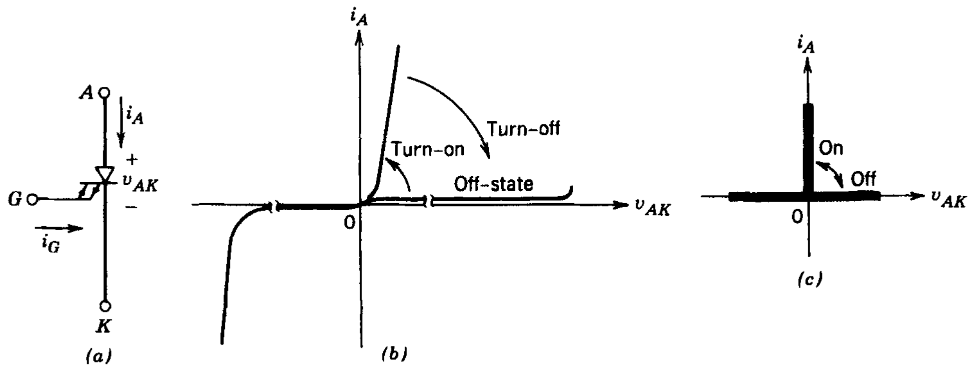

Figure 2-10 A GTO: (*a*) symbol, (*b*) *i–v* characteristics, (*c*) idealized characteristics.

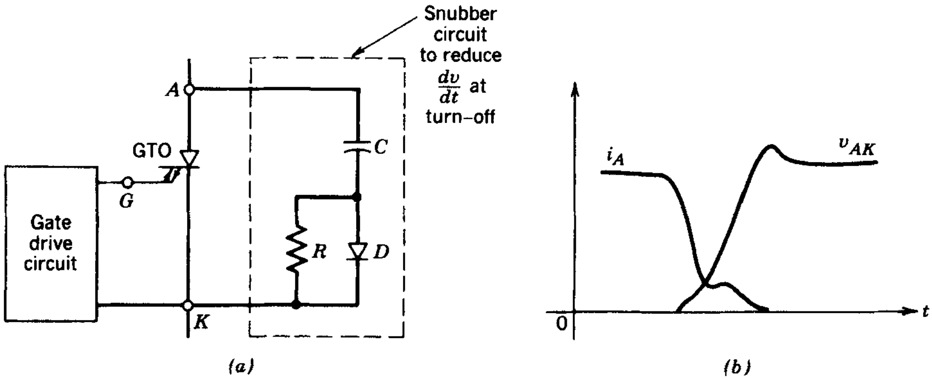

Figure 2-11 Gate turn-off transient characteristics: (*a*) snubber circuit, (*b*) GTO turn-off characteristic.

GTO design. Idealized characteristics of the device operating as a switch are shown in Fig. 2-10*c*.

Even though the GTO is a controllable switch in the same category as MOSFETs and BJTs, its turn-off switching transient is different from that shown in Fig. 2-6*b*. This is because presently available GTOs cannot be used for inductive turn-off such as is illustrated in Fig. 2-6 unless a snubber circuit is connected across the GTO (see Fig. 2-11*a*). This is a consequence of the fact that a large dv/dt that accompanies inductive turn-off cannot be tolerated by present-day GTOs. Therefore a circuit to reduce dv/dt at turn-off that consists of R, C, and D, as shown in Fig. 2-11*a*, must be used across the GTO. The resulting waveforms are shown in Fig. 2-11*b*, where dv/dt is significantly reduced compared to the dv/dt that would result without the turn-off snubber circuit. The details of designing a snubber circuit to shape the switching waveforms of GTOs are discussed in Chapter 27.

The on-state voltage (2–3 V) of a GTO is slightly higher than those of thyristors. The GTO switching speeds are in the range of a few microseconds to 25 μs. Because of their capability to handle large voltages (up to 4.5 kV) and large currents (up to a few kiloamperes), the GTO is used when a switch is needed for high voltages and large currents in a switching frequency range of a few hundred hertz to 10 kHz.

2-8 INSULATED GATE BIPOLAR TRANSISTORS

The circuit symbol for an IGBT is shown in Fig. 2-12*a* and its *i–v* characteristics are shown in Fig. 2-12*b*. The IGBTs have some of the advantages of the MOSFET, the BJT, and the GTO combined. Similar to the MOSFET, the IGBT has a high impedance gate, which requires only a small amount of energy to switch the device. Like the BJT, the IGBT has a small on-state voltage even in devices with large blocking voltage ratings (for example, V_{on} is 2–3 V in a 1000-V device). Similar to the GTO, IGBTs can be designed to block negative voltages, as their idealized switch characteristics shown in Fig. 2-12*c* indicate.

Insulated gate bipolar transistors have turn-on and turn-off times on the order of 1 μs and are available in module ratings as large as 1700 V and 1200 A. Voltage ratings of up to 2–3 kV are projected.

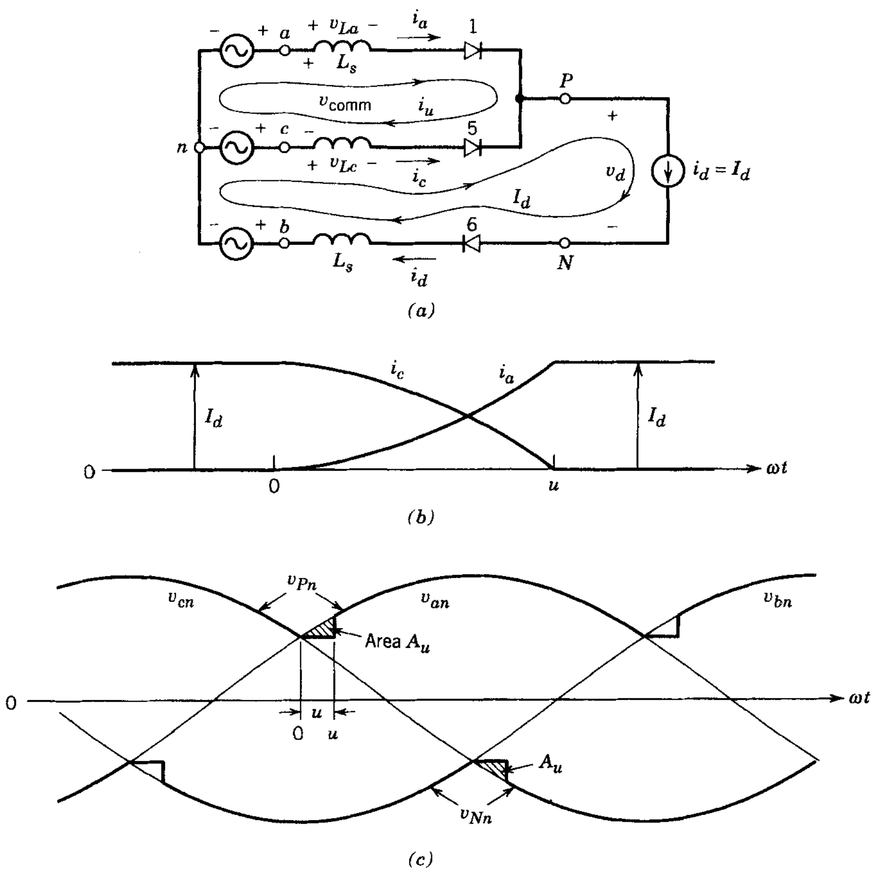

Figure 5-35 Current commutation process.

The current commutation only involves phases a and c, and the commutation voltage responsible is $v_{\text{comm}} = v_{an} - v_{cn}$. The two mesh currents i_u and I_d are labeled in Fig. 5-35a. The commutation current i_u flows due to a short-circuit path provided by the conducting diode 5. In terms of the mesh currents, the phase currents are

$$i_a = i_u$$

and

$$i_c = I_d - i_u \tag{5-74}$$

These are plotted in Fig. 5-35b, where i_u builds up from zero to I_d at the end of the commutation interval $\omega t_u = u$. In the circuit of Fig. 5-35a,

$$v_{La} = L_s \frac{di_a}{dt} = L_s di_u/dt \tag{5-75}$$

and

$$v_{Lc} = L_s \frac{di_c}{dt} = -L_s \frac{di_u}{dt} \tag{5-76}$$

noting that $i_c = I_d - i_u$ and therefore $di_c/dt = d(I_d - i_u)/dt = -di_u/dt$. Applying KVL in the upper loop in the circuit of Fig. 5-35*a* and using the above equations yield

$$v_{\text{comm}} = v_{an} - v_{cn} = v_{La} - v_{Lc} = 2L_s \frac{di_u}{dt} \tag{5-77}$$

Therefore from the above equation,

$$L_s \frac{di_u}{dt} = \frac{v_{an} - v_{cn}}{2} \tag{5-78}$$

The commutation interval u can be obtained by multiplying both sides of Eq. 5-78 by ω and integrating:

$$\omega L_s \int_0^{I_d} di_u = \int_0^u \frac{v_{an} - v_{cn}}{2} d(\omega t) \tag{5-79}$$

where the time origin is assumed to be at the beginning of the current commutation. With this choice of time origin, we can express the line-to-line voltage $(v_{an} - v_{cn})$ as

$$v_{an} - v_{cn} = \sqrt{2} V_{LL} \sin \omega t \tag{5-80}$$

Using Eq. 5-80 in Eq. 5-79 yields

$$\omega L_s \int_0^{I_d} di_u = \omega L_s I_d = \frac{\sqrt{2} V_{LL}(1 - \cos u)}{2} \tag{5-81}$$

or

$$\cos u = 1 - \frac{2\omega L_s I_d}{\sqrt{2} V_{LL}} \tag{5-82}$$

If the current commutation was instantaneous due to zero L_s, then the voltage v_{Pn} will be equal to v_{an} beginning with $\omega t = 0$, as in Fig. 5-35*c*. However, with a finite L_s, during $0 < \omega t < \omega t_u$ in Fig. 5-35*c*

$$v_{Pn} = v_{an} - L_s \frac{di_u}{dt} = \frac{v_{an} + v_{cn}}{2} \quad \text{(using Eq. 5-78)} \tag{5-83}$$

where the voltage across $L_s[=L_s(di_u/dt)]$ is the drop in the voltage v_{Pn} during the commutation interval shown in Fig. 5-35*c*. The integral of this voltage drop is the area A_u, which according to Eq. 5-81 is

$$A_u = \omega L_s I_d \tag{5-84}$$

This area is "lost" every 60°($\pi/3$ rad) interval, as shown in Fig. 5-35*c*. Therefore, the average dc voltage output is reduced from its V_{do} value, and the voltage drop due to commutation is

$$\Delta V_d = \frac{\omega L_s I_d}{\pi/3} = \frac{3}{\pi} \omega L_s I_d \tag{5-85}$$

Therefore, the average dc voltage in the presence of a finite commutation interval is

$$V_d = V_{do} - \Delta V_d = 1.35 V_{LL} - \frac{3}{\pi} \omega L_s I_d \tag{5-86}$$

where V_{do} is the average voltage with an instantaneous commutation due to $L_s = 0$, given by Eq. 5-68.

5-6-3 A CONSTANT dc-SIDE VOLTAGE $v_d(t) = V_d$

Next, we will consider the circuit shown in Fig. 5-36*a*, where the assumption is that the dc-side voltage is a constant dc. It is an approximation to the circuit of Fig. 5-30 with a large value of C. In order to simplify our analysis, we will make an assumption that the current i_d on the dc side of the rectifier flows discontinuously, and therefore only two diodes—one from the top group and one from the bottom group—conduct at any given time. This assumption allows the equivalent circuit shown in Fig. 5-36*b*, where the input voltage is made up of the line-to-line voltage segments shown in Fig. 5-36*c*. The diode D_P corresponds to one of the diodes D_1, D_3, and D_5 from the top group. Similarly, the diode D_N corresponds to one of the diodes D_2, D_4, and D_6. Similar to the analysis in Section 5-3-3 for a single-phase input, the resulting phase current waveform is shown in Fig. 5-36*c*.

5-6-3-1 Distortion in the Line-Current Waveforms

It is useful to know the power factor, total harmonic distortion, and the dc output voltage in the practical circuit of Fig. 5-30. However, to present these results in a generalized manner requires the assumption of a constant dc output voltage, as made in this section.

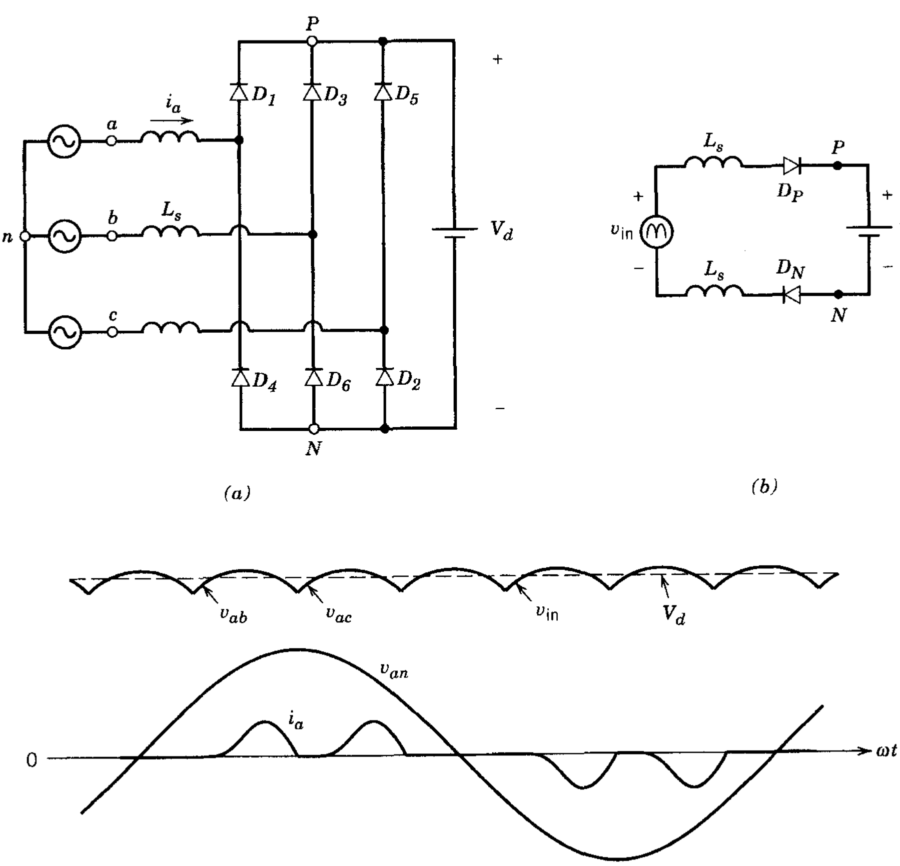

Figure 5-36 (*a*) Three-phase rectifier with a finite L_s and a constant dc voltage. (*b*) Equivalent circuit. (*c*) Waveforms.

through one of the thyristors of the top group (thyristors 1, 3, and 5) and one of the bottom group (2, 4, and 6). If the gate currents were continuously applied, the thyristors in Fig. 6-19 would behave as diodes and their operation would be similar to that described in the previous chapter. Under these conditions ($\alpha = 0$ and $L_s = 0$), the voltages and the current in phase a are shown in Fig. 6-20a. The average dc voltage V_{do} is as in Eq. 5-68:

$$V_{do} = \frac{3\sqrt{2}}{\pi} V_{LL} = 1.35 V_{LL} \tag{6-36}$$

Using the same definition as in Section 6-3-1, instants of natural conduction for the various thyristors are shown in Fig. 6-20a by 1, 2, ... The effect of the firing or delay angle α on the converter waveforms are shown in Figs. 6-20b through d. Focusing on the commutation of current from thyristor 5 to 1, we see that thyristor 5 keeps on conducting until $\omega t = \alpha$, at which instant the current commutates instantaneously to thyristor 1 due to zero L_s. The current in phase a is shown in Fig. 6-20c. Similar delay by an angle α takes place in the conduction of other thyristors. The line-to-line ac voltages and the dc output voltage v_d ($=v_{Pn} - v_{Nn}$) are shown in Fig. 6-20d.

The expression for the average dc voltage can be obtained from the waveforms in Figs. 6-20b and d. The volt-second area A_α (every 60°) results in the reduction in the average dc voltage with a delay angle α compared to V_{do} in Fig. 6-20a. Therefore,

$$V_{d\alpha} = V_{do} - \frac{A_\alpha}{\pi/3} \tag{6-37}$$

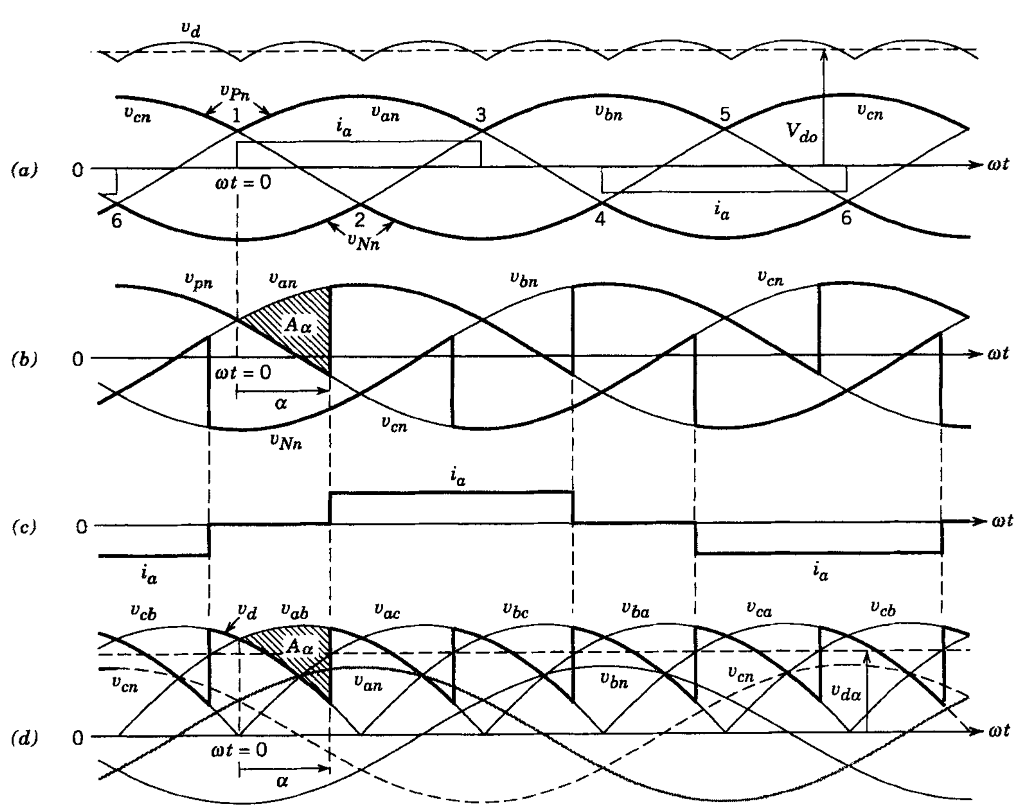

Figure 6-20 Waveforms in the converter of Fig. 6-19.

From Fig. 6-20*b*, the volt-radian area A_α is the integral of $v_{an} - v_{cn}$ $(=v_{ac})$. This can be confirmed by Fig. 6-20*d*, where A_α is the integral of $v_{ab} - v_{cb}$ $(=v_{ac})$. With the time origin chosen in Fig. 6-20,

$$v_{ac} = \sqrt{2}V_{LL}\sin\omega t \tag{6-38}$$

Therefore,

$$A_\alpha = \int_0^\alpha \sqrt{2}V_{LL}\sin\omega t\, d(\omega t) = \sqrt{2}V_{LL}(1 - \cos\alpha) \tag{6-39}$$

Substituting A_α in Eq. 6-37 and using Eq. 6-36 for V_{do} yield

$$V_{d\alpha} = \frac{3\sqrt{2}}{\pi}V_{LL}\cos\alpha = 1.35V_{LL}\cos\alpha = V_{do}\cos\alpha \tag{6-40}$$

The above procedure to obtain $v_{d\alpha}$ is straightforward when $\alpha < 60°$. For $\alpha > 60°$ we get the same result but an alternate derivation may be easier (see ref. 2).

Equation 6-40 shows that $V_{d\alpha}$ is independent of the current magnitude I_d so long as i_d flows continuously (and $L_s = 0$). The control of V_d as a function of α is similar to the single-phase case shown by Fig. 6-7. The dc voltage waveform for various values of α is shown in Fig. 6-21. The average power is

$$P = V_d I_d = 1.35V_{LL}I_d\cos\alpha \tag{6-41}$$

6-4-1-1 dc-Side Voltage

Each of the dc-side voltage waveforms shown in Fig. 6-21 consists of a dc (average) component $V_{d\alpha}$ $(=1.35V_{LL}\cos\alpha$ given by Eq. 6-40). As seen from Fig. 6-21, the ac ripple in v_d repeats at six times the line frequency. The harmonic components can be obtained by means of Fourier analysis.

6-4-1-2 Input Line Currents i_a, i_b, and i_c

The input currents i_a, i_b, and i_c have rectangular waveforms with an amplitude I_d. The waveform of i_a is phase shifted by the delay angle α in Fig. 6-22*a* with respect to its waveform in Fig. 6-20*a* with $\alpha = 0$. It can be expressed in terms of its Fourier components (with ωt defined to be zero at the positive zero crossing of v_{an}) as

$$\begin{aligned} i_a(\omega t) = {} & \sqrt{2}I_{s1}\sin(\omega t - \alpha) - \sqrt{2}I_{s5}\sin[5(\omega t - \alpha)] - \sqrt{2}I_{s7}\sin[7(\omega t - \alpha)] \\ & + \sqrt{2}I_{s11}\sin[11(\omega t - \alpha)] + \sqrt{2}I_{s13}\sin[13(\omega t - \alpha)] \\ & - \sqrt{2}I_{s17}\sin[17(\omega t - \alpha)] - \sqrt{2}I_{s19}\sin[19(\omega t - \alpha)] \cdots \end{aligned} \tag{6-42}$$

where only the nontriplen odd harmonics h are present and

$$h = 6n \pm 1 \qquad (n = 1, 2, \ldots) \tag{6-43}$$

The rms value of the fundamental-frequency component is

$$I_{s1} = 0.78I_d \tag{6-44}$$

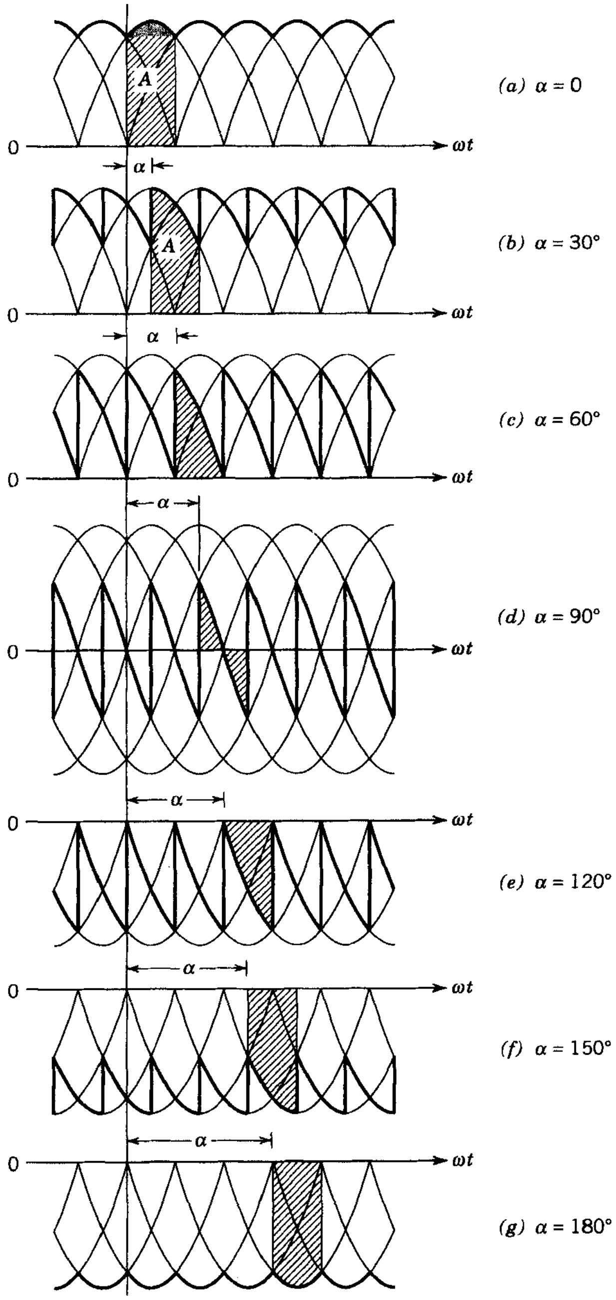

Figure 6-21 The dc-side voltage waveforms as a function of α where $V_{d\alpha} = A/(\pi/3)$. (From ref. 2 with permission.)

and the rms values of the harmonic components are inversely proportional to their harmonic order,

$$I_{sh} = \frac{I_{s1}}{h} \quad \text{where } h = 6n \pm 1 \tag{6-45}$$

as plotted in Fig. 6-22*b*.

Table 8-1 Generalized Harmonics of v_{Ao} for a Large m_f.

h \ m_a	0.2	0.4	0.6	0.8	1.0
1 *Fundamental*	0.2	0.4	0.6	0.8	1.0
m_f	1.242	1.15	1.006	0.818	0.601
$m_f \pm 2$	0.016	0.061	0.131	0.220	0.318
$m_f \pm 4$					0.018
$2m_f \pm 1$	0.190	0.326	0.370	0.314	0.181
$2m_f \pm 3$		0.024	0.071	0.139	0.212
$2m_f \pm 5$				0.013	0.033
$3m_f$	0.335	0.123	0.083	0.171	0.113
$3m_f \pm 2$	0.044	0.139	0.203	0.176	0.062
$3m_f \pm 4$		0.012	0.047	0.104	0.157
$3m_f \pm 6$				0.016	0.044
$4m_f \pm 1$	0.163	0.157	0.008	0.105	0.068
$4m_f \pm 3$	0.012	0.070	0.132	0.115	0.009
$4m_f \pm 5$			0.034	0.084	0.119
$4m_f \pm 7$				0.017	0.050

Note: $(\hat{V}_{Ao})_h/\frac{1}{2}V_d$ [$= (\hat{V}_{AN})_h/\frac{1}{2}V_d$] is tabulated as a function of m_a.

■ ***Example 8-1*** In the circuit of Fig. 8-4, $V_d = 300$ V, $m_a = 0.8$, $m_f = 39$, and the fundamental frequency is 47 Hz. Calculate the rms values of the fundamental-frequency voltage and some of the dominant harmonics in v_{Ao} using Table 8-1.

Solution From Table 8-1, the rms voltage at any value of h is given as

$$(V_{Ao})_h = \frac{1}{\sqrt{2}}\frac{V_d}{2}\frac{(\hat{V}_{Ao})_h}{V_d/2}$$

$$= 106.07\frac{(\hat{V}_{Ao})_h}{V_d/2} \tag{8-11}$$

Therefore, from Table 8-1 the rms voltages are as follows:

Fundamental:
$(V_{Ao})_1 = 106.07 \times 0.8 = 84.86\ V$ at 47 Hz
$(V_{Ao})_{37} = 106.07 \times 0.22 = 23.33\ V$ at 1739 Hz
$(V_{Ao})_{39} = 106.07 \times 0.818 = 86.76\ V$ at 1833 Hz
$(V_{Ao})_{41} = 106.07 \times 0.22 = 23.33\ V$ at 1927 Hz
$(V_{Ao})_{77} = 106.07 \times 0.314 = 33.31\ V$ at 3619 Hz
$(V_{Ao})_{79} = 106.07 \times 0.314 = 33.31\ V$ at 3713 Hz
etc. ■

Now we discuss the selection of the switching frequency and the frequency modulation ratio m_f. Because of the relative ease in filtering harmonic voltages at high frequencies, it is desirable to use as high a switching frequency as possible, except for one significant drawback: Switching losses in the inverter switches increase proportionally with the switching frequency f_s. Therefore, in most applications, the switching frequency is selected to be either less than 6 kHz or greater than 20 kHz to be above the audible range. If the optimum switching frequency (based on the overall system performance) turns out to be somewhere in the 6–20-kHz range, then the disadvantages of increasing it to 20 kHz are often outweighed by the advantage of no audible noise with f_s of 20 kHz or greater. Therefore, in 50- or 60-Hz type applications, such as ac motor drives (where the fundamental frequency of the inverter output may be required to be as high as 200 Hz), the frequency modulation ratio m_f may be 9 or even less for switching frequencies of less than 2 kHz. On the other hand, m_f will be larger than 100 for switching frequencies higher than 20 kHz. The desirable relationships between the triangular waveform signal and the control voltage signal are dictated by how large m_f is. In the discussion here, $m_f = 21$ is treated as the borderline between large and small, though its selection is somewhat arbitrary. Here, it is assumed that the amplitude modulation ratio m_a is less than 1.

8-2-1-1 Small m_f ($m_f \leq 21$)

1. *Synchronous PWM.* For small values of m_f, the triangular waveform signal and the control signal should be synchronized to each other (synchronous PWM) as shown in Fig. 8-5*a*. This synchronous PWM requires that m_f be an integer. The reason for using the synchronous PWM is that the asynchronous PWM (where m_f is not an integer) results in subharmonics (of the fundamental frequency) that are very undesirable in most applications. This implies that the triangular waveform frequency varies with the desired inverter frequency (e.g., if the inverter output frequency and hence the frequency of v_{control} is 65.42 Hz and $m_f = 15$, the triangular wave frequency should be exactly $15 \times 65.42 = 981.3$ Hz).
2. *m_f should be an odd integer.* As discussed previously, m_f should be an odd integer except in single-phase inverters with PWM unipolar voltage switching, to be discussed in Section 8-3-2-2.

8-2-1-2 Large m_f ($m_f > 21$)

The amplitudes of subharmonics due to asynchronous PWM are small at large values of m_f. Therefore, at large values of m_f, the asynchronous PWM can be used where the frequency of the triangular waveform is kept constant, whereas the frequency of v_{control} varies, resulting in noninteger values of m_f (so long as they are large). However, if the inverter is supplying a load such as an ac motor, the subharmonics at zero or close to zero frequency, even though small in amplitude, will result in large currents that will be highly undesirable. Therefore, the asynchronous PWM should be avoided.

8-2-1-3 Overmodulation ($m_a > 1.0$)

In the previous discussion, it was assumed that $m_a \leq 1.0$, corresponding to a sinusoidal PWM in the linear range. Therefore, the amplitude of the fundamental-frequency voltage varies linearly with m_a, as derived in Eq. 8-7. In this range of $m_a \leq 1.0$, PWM pushes the harmonics into a high-frequency range around the switching frequency and its multiples. In spite of this desirable feature of a sinusoidal PWM in the linear range, one of the drawbacks is that the maximum available amplitude of the fundamental-frequency component is not as high as we wish. This is a natural consequence of the notches in the output voltage waveform of Fig. 8-5*b*.

To increase further the amplitude of the fundamental-frequency component in the output voltage, m_a is increased beyond 1.0, resulting in what is called *overmodulation*. Overmodulation causes the output voltage to contain many more harmonics in the sidebands as compared with the linear range (with $m_a \leq 1.0$), as shown in Fig. 8-7. The harmonics with dominant amplitudes in the linear range may not be dominant during